EARLY STAGE FEEDBACK EVALUATION FOR CONTINUOUS IMPROVEMENT

Goal: To gather data space experts and/or practitioners feedback on the proposed framework and the current proof-of-concept prototype in order to support the continued development and future evaluation of the framework.

Method: Given the early stage of the proposed framework, we propose an informal workshop where a member of the research team presents the current results: an overview of the proposed framework and the proof-of-concept prototype.

Protocol and instruments:

- Presentation of an overview of the proposed framework and the current proof-of-concept prototype.(20 mins + time for questions)
- Questionnaire to gather feedback (early feedback form, see below).
- Content analysis of the received feedback.

Intended Results:

- Insights to improve the suggested framework.

Early Feedback Form

Name (optional): María

Once you have been provided with an overview of the proposed framework, please answer the following:

1) Write the three aspects you like the most of the suggested framework and the proof-of-concept prototype and three aspects you would like to improve/change/eliminate and explain the reason.

Aspects you like the most of the suggested framework and the proof-of-concept prototype	**Explanation**
Alignment with a common policy language.	Expressing data quality requirements in ODRL keeps them within the same policy language used in the data space, rather than introducing a separate representation for quality conditions.
Relevance to a practical problem.	The framework addresses a concrete gap in data space projects: the need to manage data quality requirements and how to connect them with implementation.
End-to-end coverage.	The framework and prototype connect the definition of DQRs with their implementation as validation checks, rather than leaving the transition between these stages unresolved.

Aspects you would like to improve/change/eliminate	**Explanation**
Avoid the proliferation of nearly identical patterns.	Small differences between patterns could be made the framework hard to use.
Reduce dependence on the central authority.	Responsibility for defining and applying DQRs should be distributed more clearly among participants. The authority should coordinate data quality governance without being expected to perform every task itself. In this sense, the authority may have an approval and assurance role. The authority should understand and review what DQRs providers define and implement, and give an endorsement or quality seal where appropriate. This role could also be performed by an external body acting under delegated authority.

2) Do you have additional comments?

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Early Feedback Form

Name (optional): Lucía

Once you have been provided with an overview of the proposed framework, please answer the following:

3) Write the three aspects you like the most of the suggested framework and the proof-of-concept prototype and three aspects you would like to improve/change/eliminate and explain the reason.

Aspects you like the most of the suggested framework and the proof-of-concept prototype	**Explanation**
Relevance to a practical problem.	The framework addresses a concrete gap in data space projects: the need to manage data quality requirements and how to connect them with implementation.
End-to-end coverage.	The framework and prototype connect the definition of DQRs with their implementation as validation checks, rather than leaving the transition between these stages unresolved.
Support for explicit quality commitments.	The approach provides a basis for making quality conditions explicit (and understandable) between providers and consumers. This is especially relevant when moving towards paid access to datasets and including quality commitments in data-sharing agreements.

Aspects you would like to improve/change/eliminate	**Explanation**
Give providers a stronger role in defining specialized requirements.	The governance authority cannot be expected to have detailed knowledge of every dataset. Providers should contribute DQRs, while the authority reviews and approves them within the common governance arrangements.
Reduce dependence on the central authority.	Responsibility for defining and applying DQRs should be distributed more clearly among participants. The authority should coordinate data quality governance without being expected to perform every task itself. In this sense, the authority may have an approval and assurance role. The authority should understand and review what DQRs providers define and implement, and give an endorsement or quality seal where appropriate. This role could also be performed by an external body acting under delegated authority.
Accommodate consumer-specific quality needs.	Consumers could have a way to request quality requirements that the authority has not initially anticipated. These requests should be considered through the common governance process rather than being excluded by a fixed set of centrally defined requirements.

4) Do you have additional comments?

It is an interesting approach for exploring how to make connections between contractual conditions and implementation. i.e., the end-to-end approach offers a way to connect quality requirements with their technical validation, going beyond the option of simply writing those conditions as DQRs but creating data-sharing contracts with data quality clauses. Also, it is related with value for dataset monetization. Making quality requirements explicit is particularly important when datasets are monetized, because providers and consumers need a clear understanding of the quality conditions associated with the data being offered.

Early Feedback Form

Name (optional): Adrián

Once you have been provided with an overview of the proposed framework, please answer the following:

5) Write the three aspects you like the most of the suggested framework and the proof-of-concept prototype and three aspects you would like to improve/change/eliminate and explain the reason.

Aspects you like the most of the suggested framework and the proof-of-concept prototype	**Explanation**
Relevance to a practical problem.	The framework addresses a concrete gap in data space projects: the need to manage data quality requirements and how to connect them with implementation.
Support for multiple data columns.	The ability to express requirements involving several columns is valuable, as some quality conditions concern relationships between attributes rather than individual values considered in isolation.
Alignment with a common policy language.	Expressing data quality requirements in ODRL keeps them within the same policy language used in the data space, rather than introducing a separate representation for quality conditions.

Aspects you would like to improve/change/eliminate	**Explanation**
Prevent excessive fragmentation of data models.	Customization could become so fine-grained that a separate data model is needed for each dataset. The framework should address this risk and explain how shared models and reusable requirements can be maintained.
Reduce dependence on the central authority.	Responsibility for defining and applying DQRs should be distributed more clearly among participants. The authority should coordinate data quality governance without being expected to perform every task itself. In this sense, the authority may have an approval and assurance role. The authority should understand and review what DQRs providers define and implement, and give an endorsement or quality seal where appropriate. This role could also be performed by an external body acting under delegated authority.

6) Do you have additional comments?

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Early Feedback Form

Name (optional): Alberto

Once you have been provided with an overview of the proposed framework, please answer the following:

7) Write the three aspects you like the most of the suggested framework and the proof-of-concept prototype and three aspects you would like to improve/change/eliminate and explain the reason.

Aspects you like the most of the suggested framework and the proof-of-concept prototype	**Explanation**
Relevance to a practical problem.	The framework addresses a concrete gap in data space projects: the need to specify data quality requirements and how to connect them with implementation.
End-to-end coverage.	The framework and prototype connect the definition of DQRs with their implementation as validation checks, rather than leaving the transition between these stages unresolved.
Support for multiple data columns.	The ability to express requirements involving several columns is valuable, as some quality conditions concern relationships between attributes rather than individual values considered in isolation.

Aspects you would like to improve/change/eliminate	**Explanation**
Give providers a stronger role in defining specialized requirements.	The governance authority cannot be expected to have detailed knowledge of every dataset. Providers should contribute DQRs, while the authority reviews and approves them within the common governance arrangements.

8) Do you have additional comments?

An interesting application would be quality assurance for sensor data, considering differences in measurement error, precision, and units. The framework could support assessing whether sensors and their outputs meet agreed requirements, providing greater confidence to sensor buyers and data consumers, including in sensor-as-a-service arrangements. A possible connection with the Data Act would also be worth exploring.